Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/07/2021
PI Name:	Smith, David J Ph.D.		
Project Title:	Accumulated Doses and Microbial Mutagenesis from Ionizing Radiation Exposures (ADMMIRE) over Antarctica		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-4819
Organization Name:	NASA Ames Research Center		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	10/01/2018	**End Date:**	05/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: Project concluded in May 2021 (Original end date was 9/30/2021), per PI and Y. Zhang/KSC (Ed., 7/9/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gentry, Diana Ph.D. (NASA Ames Research Center) Venkateswaran, Kasthuri Ph.D. (Jet Propulsion Laboratory)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	How bacteria respond to low dose rate (LDR) ionizing radiation is poorly understood because such experiments are often impractical at ground-based research facilities. Alarmingly, previous spaceflight studies have hinted at evolved bacterial virulence and emerging novel species in spacecraft. It is conceivable that the radiation quality in space, including high energy ions and protons, were responsible for these observations; however, orbital spaceflight experiments include both microgravity and high energy radiation exposures (simultaneously) making it hard to untangle the influence of each stressor. Balloons launched to the polar stratosphere can be used to perform studies capable of measuring the influence of LDR ionizing radiation exposures, alone, on microorganisms. Our study, "Accumulated Doses and Microbial Mutagenesis from Ionizing Radiation Exposures (ADMIRRE) over Antarctica," will use the Exposing Microorganisms in the Stratosphere (E-MIST) payload flown on a long duration polar balloon mission. Previous surveys of the International Space Station (ISS) have revealed that most viable environmental bacteria are dormant, accumulating radiation over time but not actively growing on surfaces. How does prolonged dormancy in the space radiation environment impact the behavior of bacteria when opportunistically resuscitated? Our proposed experiment can address this major unknown with four achievable objectives: (1) Prepare the E-MIST payload and bacterial spores for a long-duration balloon flight; (2) Fly the experiment to the Antarctic stratosphere and measure environmental conditions; (3) Determine the effect of near space polar radiation on bacterial samples; and (4) Simulate near space radiation conditions using ground-based facilities. Our overarching project hypothesis is that accumulated LDR ionizing radiation will result in measurable, non-lethal mutations in dormant Bacillus siamensis IIF7SW-P1 spores. The E-MIST payload was designed by our team and has flown multiple times to the stratosphere as a NASA piggyback payload, including a brief 8-hour Antarctic flight launched in December 2018, and another 32-day Antarctic flight launched in December 2019. For this proposed experiment, we will work with ISS-isolated bacteria collected from Space Biology's Microbial Tracking-1 project. The bacterial specimens will be sporulated to create dormant, uniform, desiccated and stable samples; thus, our strains are ideal for Antarctic mission logistics that entail long handover and retrieval times. Pre-flight ground chamber testing, hardware refurbishment, and instrument calibrations will be done at Kennedy Space Center (KSC) by members of our team. The E-MIST payload will include a Technology Readiness Level-9 (TRL-9) dosimeter for measuring the radiation environment; additionally, we will insert shielding (aluminum, titanium, and polypropylene) with varying thicknesses 'upstream' of bacterial sample plates in order to determine the effect of radiation absorbance and/or particle scattering. Pre- and post-flight microbiological processes will be done at Ames Research Center (ARC) and Jet Propulsion Laboratory (JPL), where surviving bacteria will be germinated and characterized using state-of-the-art molecular assays. Project omics data (RNASeq differential expression analysis) will be archived in the NASA GeneLab repository. By studying the response and potential mutagenesis of dormant bacteria in space-like radiation conditions, our project is directly aligned with priorities outlined in the Space Biology Science Plan 2016-2025 and Decadal Survey, specifically: MB-1 (What underlying genetic, molecular and biochemical processes are influenced by the space environment?) /Decadal P1 (multigenerational studies of microbial populations), P2 (analysis in space of plant and microbial growth and of the physiological responses of plants and microbes to the spaceflight environment), and MB-6 (Does long-duration spaceflight alter normal rates of evolutionary change?) /Decadal P1. Using the NASA balloons to deliver a natural, complex, LDR ionizing radiation exposure in the polar stratosphere will allow our team to better understand how spacecraft-associated bacteria could be influenced by radiation levels expected on a long duration mission to Mars, and if any changes pose a potential risk to astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project seeks to distinguish between acute and LDR radiation exposure effects on bacterial spores, with measurements of antibiotic resistance (and associated genes) in surviving specimens. Antibiotic resistance is a growing public health threat on Earth. A deeper understanding of antibiotic resistance profiles -- regardless of the trigger (in our case: ionizing radiation) -- will be useful to the scientific community searching for solutions.
Task Progress:	FINAL REPORTING -- JULY 2021 Due to COVID-19 associated delays across NASA and our science team’s laboratories, key team personnel changes, and Balloon Program Office scheduling constraints, we have concluded the ADMMIRE project, effective May 2021. Some publications have been submitted and are pending publication. ANNUAL REPORTING -- JULY 2020 Due to COVID-19 associated delays across NASA and our science team’s laboratories, we have “paused” the ADMMIRE project, effective June 2020. This decision was made after discussions with Balloon Program Office and indications that another flight opportunity from Antarctica for the ADMMIRE project would be unlikely to occur until FY22. Up until this point, however, major ADMMIRE project advancements have been made, including new balloon flight results from FY19/20 missions flown from New Mexico and Antarctica. A short summary of FY20 milestones are provided below. Task 1: Refurbishment and testing of E-MIST hardware & instruments = Complete. Payload refurbished at KSC after 32-day mission flown from Antarctica (Planetary Protection research grant). Task 2: Machine coupons of varying thicknesses for carrying spores = In-Work. Aluminum coupons ready. Polypropylene and titanium coupons have not yet been manufactured by KSC Prototype Laboratory. Task 3: Prepare bacterial spores = Complete. Bacillus siamensis strains from MT-1 available and ready for use. Environmental permit extended through March 2023 for future ADMMIRE flight. Task 4: Maintain ground and transport controls = Not Yet Started. Bacillus siamensis controls will be produced prior to ground and flight experiments. Task 5: Record environmental conditions and radiation doses = Ready. We have made major strides improving the E-MIST payload radiometer and dosimeter capabilities, through citizen scientist collaborations and international partnerships with the German Aerospace Center (DLR) (M42 dosimeter). Dosimeters tested on FY19 Ft. Sumner NASA Balloon Flight (LDB Test Flight) with the Columbia Scientific Balloon Facility (CSBF) in September 2019 and Antarctic flight in Dec 2019-Jan 2020.

	Task 6: Survival of bacterial samples = In-Work. MARSBOx manuscript submitted to Astrobiology in July 2020: Cortesão et al. MARSBOx: Fungal and bacterial endurance from a balloon-flown analog mission in the stratosphere. Task 7: Antibiotic resistance tests for spores surviving radiation = In-Work. MARSBOx manuscript submitted to Astrobiology in July 2020: Cortesão et al. MARSBOx: Fungal and bacterial endurance from a balloon-flown analog mission in the stratosphere. Task 8: Differential expression for spores surviving radiation exposures = Not Yet Started. Will be performed in upcoming project years. Task 9: Low linear energy transfer (LET) control studies testing bacterial spore resistance = Complete. We have run x-ray studies at Ames to optimize bacterial survival and sequencing assays, in prep for future ground- and flight-based experiments. Task 10: Accelerated iron ion studies testing bacterial spore resistance = Not Yet Started. Will be performed in upcoming project years. Awarded “piggyback” beam time at Heavy Ion Medical Accelerator in Chiba (HIMAC, Chiba, Japan).
Bibliography Type:	Description: (Last Updated: 12/09/2021)
Articles in Other Journals or Periodicals	Berger T, Marsalek K, Przybyla B, Matthiä D, Aeckerlein J, Rohde M, Wirtz M, Möller R, James L, Lane M, Johnson P, Sowa M, Smith DJ. "On the radiation environment during consecutive balloon flights over New Mexico and Antarctica." Space Weather (in revision May 2021). Published Online: Mon, 1 Mar 2021. https://doi.org/10.1002/essoar.10506355.1 in Earth and Space Science Open Archive preprint site. , Mar-2021
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